

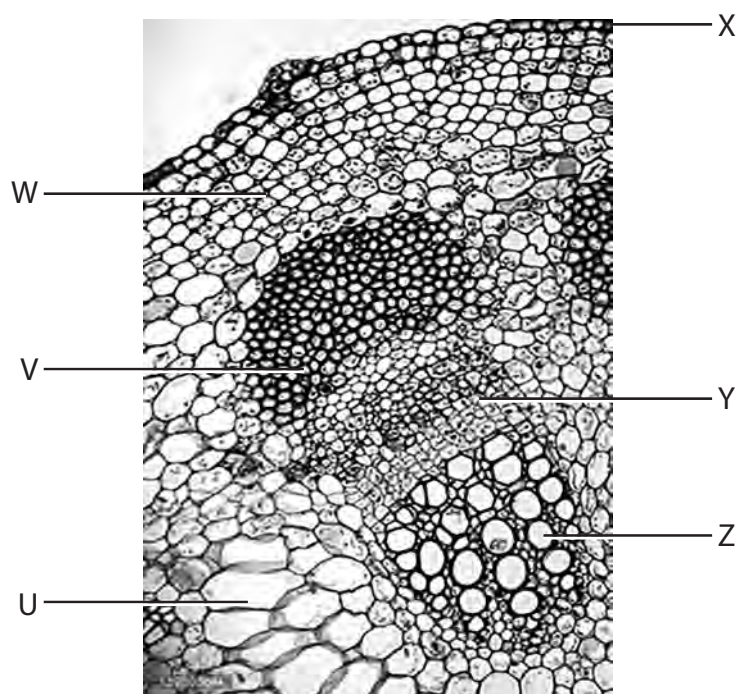
Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1** Mineral ions are transported by xylem vessels.

The photograph shows a section through part of the stem of a plant, as seen using a light microscope.



- (a) (i) Which letter identifies a xylem vessel?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

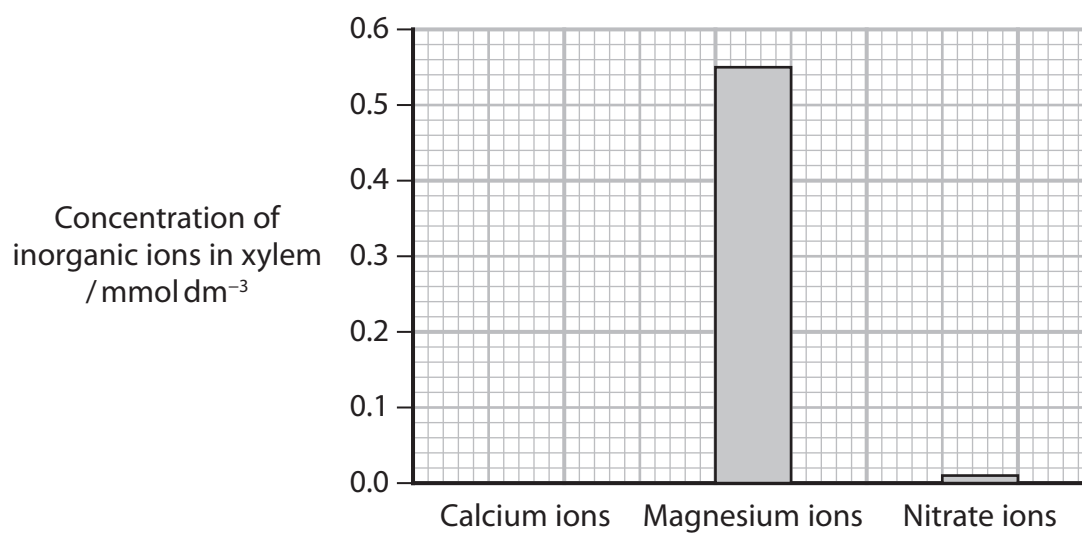
- (ii) Which letter identifies a sclerenchyma fibre?

(1)

- ☐ **A** U
- ☐ **B** V
- ☐ **C** W
- ☐ **D** Z



(b) The graph shows the concentrations of three inorganic ions in the xylem of a plant.



(i) The concentration of calcium ions is $0.45 \text{ mmol dm}^{-3}$.

Plot this concentration on the graph.

(1)

(ii) Explain the importance of each of these ions to the plant.

(3)

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(Total for Question 1 = 6 marks)



P 6 1 4 7 0 A 0 3 3 2